1. 解: $dX = kXdt + \sigma dB_t$, $Y(t) = X(t, B_t)e^{kt} = g(X, t)$

$$dY = \frac{\partial g}{\partial X}dX + \frac{\partial g}{\partial t}dt + \frac{1}{2}\frac{\partial^2 g}{\partial X^2}(dX)^2$$

$$= e^{kt}dX + kXe^{kt}dt + \frac{1}{2}\times 0$$

$$= e^{kt}(kXdt + \sigma dB_t + kXdt) = 2kXe^{kt}dt + \sigma e^{kt}dB_t$$

复习: $\int_0^t f(s)dB_s \sim N(0, \int_0^t f^2(s)ds)$ [基本的性质]

2. 证明: $\frac{dS}{S} = \mu dt + \sigma dB_t$, 远期价格: $F_t = S_t e^{r(T-t)}$

则 $dF_t = e^{r(T-t)}dS_t - rS_t e^{r(T-t)}dt$

$$= e^{r(T-t)}(\mu S_t dt + \sigma S_t dB_t - rS_t dt)$$

$$= e^{r(T-t)}S_t((\mu - r)dt + \sigma dB_t)$$

$\Rightarrow \frac{dF_t}{F_t} = (\mu - r)dt + \sigma dB_t$ 则 F_t 遵循几何Brownian运动

3. Sol: 年收益率 $r \sim (0.16, 0.3)$, $\mu = 0.16$, $\sigma = 0.3$

$\ln S_t = \ln S_0 + (\mu - \frac{\sigma^2}{2})t + \sigma B_t$, $S_0 = \$50$ 复习对数正态分布的n阶矩

(1) $E(S_{T=\frac{1}{360}}) = S_0 e^{\mu T} = \$50\exp(\frac{0.16}{360})$ 漏题, 还有 $Var(S_{\frac{1}{360}})$, 用n阶矩即可

注意"年"收益和"天"的关系

(2) $P(S_1 > 52) = P(\ln S_1 > \ln 52)$ $\ln S_1 \sim N(\ln 50 + \frac{(0.16 - \frac{0.09}{2})}{360}, \frac{0.09}{360})$

$= N(\ln 50 + \frac{0.115}{360}, \frac{0.09}{360})$

$$= P\left(\frac{\ln S_1 - (\ln 50 + \frac{0.115}{360})}{0.3/\sqrt{360}} > \frac{\ln 52 - \ln 50 + \frac{0.115}{360}}{0.3/\sqrt{360}}\right)$$

$$= \Phi\left(\frac{\ln\frac{50}{52} - \frac{0.115}{360}}{0.3/\sqrt{360}}\right) = \Phi\left(\frac{\ln\frac{50}{52} - \frac{0.115}{360}}{1/20\sqrt{10}}\right) = \Phi\left(20\sqrt{10}\left(\ln\frac{50}{52} - \frac{0.115}{360}\right)\right)$$

(3) $t=30$ 提前执行的条件: $C_{\frac{30}{360}} < X - S_{\frac{30}{360}}$ 内在价值 > 价格

$\max(53 - S_{\frac{30}{360}}, 0) > 1.5$

$$\ln S_t \sim N\left(\ln S_0 + \left(\mu - \frac{\sigma^2}{2}\right)t, \sigma^2 t\right)$$
$$= N\left(\ln 50 + 0.115t, (0.3)^2 t\right)$$

参照(2)的算法:

$$P(\underbrace{C_{30}}_{X} < X - S_{30}) = P(\underbrace{S_{30}}_{X} < X - C_{30})$$
$$= \Phi\left(\frac{\ln\frac{X - C_{30}}{S_0} + 0.115 \times \frac{30}{360}}{0.3\sqrt{30}/\sqrt{360}}\right) = \Phi\left(\frac{\ln\left(\frac{51.5}{50}\right) + \frac{0.115}{12}}{\sqrt{3}/20}\right)$$

(4) 一天内 ~~收益率~~ $r \sim N\left(\frac{\mu}{360}, \frac{(0.3)^2}{360}\right) = N(0.16, 0.3^2)$

则 $P(r>0) = \left(\frac{r - 0.16}{0.3/\sqrt{360}} > \frac{-0.16}{0.3/\sqrt{360}}\right)$

$$= \Phi\left(\frac{0.16}{0.3/\sqrt{360}}\right) = \Phi(3.2\sqrt{10})$$

而两天内的收益率相互独立

则股票连涨两天的概率为 $\Phi^2(3.2\sqrt{10})$

$$\ln S_T = \ln S_t + \frac{0.115(T-t)}{360} + 0.3(B_T - B_t)$$

则 $\ln S_T - \ln S_t = \frac{0.115}{360}(T-t) + 0.3(B_T - B_t) \sim$ 正态分布

$\ln S_T - \ln S_t > 0 \Leftrightarrow S_T - S_t > 0$ 则可求

年

$r \sim N(\mu, \sigma^2)$

日 $r' \sim (\frac{\mu}{360}$